Source A · NHS public information

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Cystic fibrosis is inherited. Altered gene copies from both parents cause thick, sticky mucus. This can affect the lungs and digestion. Treatments can help ease symptoms. Care depends on the person.

This source gives a public overview. It does not tell us one person’s exact symptoms or future.

Publisher: NHS. Title: Cystic fibrosis. Use the Causes, Symptoms and Treatments sections.

Source A full page: ask an adult to use the link in the page tools or staff notes.

Accessed 24 October 2026. Page reviewed 28 February 2025. This is a classroom paraphrase, not a quotation. The full page can be opened by the teacher.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Source B · NHS Genomics Education

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Cystic fibrosis usually involves harmful variants in both copies of CFTR. This is an autosomal recessive condition. A carrier has one changed copy in this model. The effects can vary.

Two carrier parents: each pregnancy	Model chance
Child has the condition	1 in 4 = 25%
Child is a carrier	2 in 4 = 50%
Child inherits neither changed copy	1 in 4 = 25%

These chances apply each time. They do not predict a set family count or the severity of one person’s symptoms.

NHS Genomics Education Programme. Page: Cystic fibrosis. Read Genomics, then Inheritance and genomic counselling. Use these parts to check the gene and how changed copies pass to a child.

Source B full page: ask an adult to use the link in the page tools or staff notes.

Checked 24 October 2026. This card uses our own words. It is not a quote. We have not given a review date for this source.

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Sources · purpose and limits

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Both sources are NHS sources. They have different readers and are not wholly independent publishers.

Source A	Source B
Public information about effects and care	More detail about genes and inheritance

Both support inherited cause. More detail does not by itself mean a conflict. Check the actual claim, source section, date and purpose.

Read the source, cover it, explain the meaning in your own words, then check it. Keep the source label beside each claim.

Use fictional cases. Do not ask for family details or predict a person’s future. Inheritance and infection precautions are separate matters.

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Look at the NHS source card. Point to one detail you notice.

Cystic fibrosis is inherited. It can affect breathing and digestion. Changed CFTR copies come from both parents. One changed copy usually means a carrier. Treatments can ease symptoms.

Source A and Source B are named in the pack. The genetic cause is inherited. Infection precautions are separate. This card predicts nothing about a person.

Point, sign, speak, draw, type or write. You may pass or pause. An adult can read or scribe your exact words. Plan in any language.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Starter

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Think back. Show one answer at a time.

1. What does a Punnett square show?
2. What is a gene?
3. Where are chromosomes found in an animal cell?

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Word page

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Word	Meaning	Example
inherited	passed to a child through genes	The card says CF is inherited.
variant	a changed version of a gene	Some CFTR variants cause CF.
carrier	one changed copy in this recessive model	A carrier usually does not have CF.
source	where a claim comes from	Name the NHS source beside the claim.

Use your own language to plan or explain the words.

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

I do · a source-linked model

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

The NHS card says CF is inherited. Changed CFTR copies come from both parents. One changed copy usually means a carrier. Two usually cause the condition. Name the NHS source with each claim.

Carrier F/f × carrier F/f	F	f
F	FF	Ff
f	fF	ff

F means the usual CFTR copy. f means a changed copy in this model. Ff and fF are carriers. ff is one of four possible pairs each time.

Show me: is the genetic cause inherited or passed by brief contact? Give the clue from the card.

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do · the missing steps

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

The NHS card says CF is ___. It involves the ___ gene. One changed copy usually means a ___. Source: ___.

The card names ___. This sentence supports ___. This line is about ___.

Supported: point to the condition, inheritance and care lines.

Independent: explain which line supports inheritance and which describes care.

Aspire: use the card to reject the claim that every carrier has symptoms.

One partner explains. The other asks for the clue. Swap roles. Keep your own first answer, feedback and next step.

Take an agreed one-minute reset within this stage.

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do · match a claim to its source section

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Statement	Your section
Can affect breathing and digestion	_____
Usually two changed CFTR copies	_____
Treatments help manage symptoms	_____
NHS page named and dated	_____

Choose What it is, Inheritance, Care or Source. Give the clue for each match.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do 2 · check the claim

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

A fictional family-source card says altered gene copies passed from parents to their child. Which claim does that evidence support?

A. The altered copies spread through a shared cough.

B. The altered copies arose from the child’s wish.

C. The altered copies were inherited from the parents.

D. The altered copies travelled through drinking water.

Choose for yourself. Give a science reason. Point or use a card if you wish.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Supported

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Present three sourced facts from the offline NHS card, then correct an unsupported claim.

State the condition and one fact with the NHS source named.

The card says ___. My source is ___.

Say it before you write. You may use any language to plan. Keep your own response.

Choose one route for the full 8 minutes. The one-minute reset is inside this time.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Independent

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Present three sourced facts from the offline NHS card, then correct an unsupported claim.

Explain CFTR inheritance and separate it from infection.

The CFTR change is inherited because ___. Infection advice is separate because ___.

Say it before you write. You may use any language to plan. Keep your own response.

Choose one route for the full 8 minutes. The one-minute reset is inside this time.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Aspire

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Present three sourced facts from the offline NHS card, then correct an unsupported claim.

Revise an unsupported claim with the NHS card.

The claim is not supported by ___. I would revise it to ___.

Say it before you write. You may use any language to plan. Keep your own response.

Choose one route for the full 8 minutes. The one-minute reset is inside this time.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

My sourced facts and repair

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

My fact	Source and supporting line
1. _____	_____
2. _____	_____
3. _____	_____

Unsupported claim: _____

My repair and source: _____

Actual feedback and my next step: _____

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Go deeper · a claim audit

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

A draft says a CFTR change spreads by touch and that outcomes are fixed. Repair both claims. Give the source for each repair.

Why does one changed copy not justify saying that a carrier has the condition?

Use this task within the same 8-minute budget.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Review and improve

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Keep your first answer. Hear what the adult noticed. Make one real change, or give a reason to keep it. Ask for a read-back if it helps.

Mark a real edit // and ask for a read-back ? when it helps. Take an agreed one-minute reset within this stage.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Supported

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Name the condition and source. Show the inheritance model and give one reason.

My actual feedback: _____

My next step: put the NHS source beside each science claim.

A fact I can support: _____

A claim I repaired: _____

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Independent

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Name the condition and source. Show the inheritance model and give one reason.

My actual feedback: _____

My next step: put the NHS source beside each science claim.

A fact I can support: _____

A claim I repaired: _____

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Aspire

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

Name the condition and source. Show the inheritance model and give one reason.

My actual feedback: _____

My next step: put the NHS source beside each science claim.

A fact I can support: _____

A claim I repaired: _____

Made from the web lesson SCI-LAUNCH-A2-W06
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Knowledge organiser

I can name the condition and source.

I can explain CFTR inheritance without personal details.

I can tell a sourced fact from a guess.

I am learning to use a named source to present careful facts about a genetic condition.

What the card supports	A careful limit
CF involves inherited CFTR variants	A carrier usually has one changed copy
CF can affect breathing and digestion	Effects differ between people
Treatments can ease symptoms	Do not predict a person’s future

Name the NHS source beside each fact. Keep inheritance separate from infection precautions.

Two carriers: FF, Ff, fF, ff are four possible pairs per pregnancy. The ff chance is 1 in 4. This is a model, not a fixed family count.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale